

DIFFERENTIAL EQUATIONS

Lecture 7 — Linear ODEs & Bernoulli Equations

Linear First-Order ODE

Definition: Linear First-Order ODE (Standard Form)

$$\frac{dy}{dx} + P(x)y = Q(x)$$

- If $Q(x) = 0$: the equation is **homogeneous** (not to be confused with the homogeneous substitution type from Lecture 4).
- If $Q(x) \neq 0$: the equation is **non-homogeneous**.

$P(x)$ and $Q(x)$ are functions of x only — there are no powers of y , no products of y with itself, and no transcendental functions of y . This is what makes it linear.

Getting to standard form: The equation is not always handed to you in standard form. The first thing to do is always divide through by whatever is multiplying dy/dx , to make its coefficient exactly 1. Only then can you correctly read off $P(x)$ and $Q(x)$.

Method of Solution — Integrating Factor

Solving a Linear First-Order ODE

1. Write in standard form: $\frac{dy}{dx} + P(x)y = Q(x)$.
2. Find the integrating factor: $\text{IF} = e^{\int P(x) dx}$.
3. Multiply both sides by IF. The left-hand side collapses into a single exact derivative:

$$\frac{d}{dx} [\text{IF} \cdot y] = \text{IF} \cdot Q(x)$$

4. Integrate both sides with respect to x :

$$\text{IF} \cdot y = \int [\text{IF} \cdot Q(x)] dx + C$$

5. Divide through by IF to solve for y .

Why does the LHS collapse? When you multiply $dy/dx + P(x)y$ by $\text{IF} = e^{\int P dx}$, the result is exactly the product rule expansion of $\frac{d}{dx} [\text{IF} \cdot y]$:

$$\frac{d}{dx} [\text{IF} \cdot y] = \text{IF} \cdot \frac{dy}{dx} + \text{IF} \cdot P(x) \cdot y$$

This is why we can immediately write it as a single derivative and integrate directly. The IF is specifically chosen to make this happen.

Worked Examples — Linear ODE

Example 1: $\frac{dy}{dx} + \frac{3}{x}y = 6x^2$

Step 1. Already in standard form. Read off $P(x) = 3/x$ and $Q(x) = 6x^2$.

Step 2. Find the IF:

$$\text{IF} = e^{\int 3/x \, dx} = e^{3 \ln x} = e^{\ln x^3} = x^3$$

Step 3. Multiply both sides by x^3 :

$$x^3 \frac{dy}{dx} + 3x^2 y = 6x^5$$

The LHS is $\frac{d}{dx}(x^3 y)$:

$$\frac{d}{dx}(x^3 y) = 6x^5$$

Step 4. Integrate both sides:

$$x^3 y = \int 6x^5 \, dx = x^6 + C$$

Step 5. Solve for y :

$$y = x^3 + Cx^{-3}$$

Example 2: $x \frac{dy}{dx} = 2y + x^3 e^x$

Step 1. Get to standard form. Divide both sides by x :

$$\frac{dy}{dx} = \frac{2y}{x} + x^2 e^x \implies \frac{dy}{dx} - \frac{2}{x}y = x^2 e^x$$

Now $P(x) = -2/x$ and $Q(x) = x^2 e^x$.

Step 2. Find the IF:

$$\text{IF} = e^{\int -2/x \, dx} = e^{-2 \ln x} = e^{\ln x^{-2}} = x^{-2} = \frac{1}{x^2}$$

Step 3. Multiply both sides by $1/x^2$:

$$\frac{1}{x^2} \frac{dy}{dx} - \frac{2}{x^3} y = e^x$$

The LHS is $\frac{d}{dx}\left(\frac{y}{x^2}\right)$:

$$\frac{d}{dx}\left(\frac{y}{x^2}\right) = e^x$$

Step 4. Integrate both sides:

$$\frac{y}{x^2} = \int e^x \, dx = e^x + C$$

Step 5. Solve for y :

$$y = x^2 e^x + Cx^2$$

Example 3: $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$

Step 1. Get to standard form. Divide by $(1 + x^2)$:

$$\frac{dy}{dx} + \frac{2x}{1 + x^2} y = \frac{4x^2}{1 + x^2}$$

So $P(x) = \frac{2x}{1 + x^2}$ and $Q(x) = \frac{4x^2}{1 + x^2}$.

Step 2. Find the IF:

$$\text{IF} = e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = 1 + x^2$$

Step 3. Multiply both sides by $(1 + x^2)$:

$$(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$$

The LHS is $\frac{d}{dx}[(1 + x^2)y]$:

$$\frac{d}{dx}[(1 + x^2)y] = 4x^2$$

Step 4. Integrate both sides:

$$(1 + x^2)y = \int 4x^2 dx = \frac{4x^3}{3} + C$$

Step 5. Solve for y :

$$y = \frac{4x^3}{3(1 + x^2)} + \frac{C}{1 + x^2}$$

Bernoulli Equation

Definition: Bernoulli Equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \neq 0, 1$$

This looks like a linear ODE but has a y^n on the right-hand side, making it **non-linear**. The trick is to reduce it to a linear ODE using a substitution.

When $n = 0$ the equation is already linear (the RHS is just $Q(x)$). When $n = 1$, the equation is separable. The Bernoulli method is needed for all other values of n .

Reduction to a Linear ODE

Solving a Bernoulli Equation

Step 1. Divide both sides by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Step II. Substitute $v = y^{1-n}$. Differentiating: $\frac{dv}{dx} = (1-n) y^{-n} \frac{dy}{dx}$, so:

$$y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx}$$

The equation becomes linear in v :

$$\frac{1}{1-n} \frac{dv}{dx} + P(x) v = Q(x) \quad \implies \quad \frac{dv}{dx} + (1-n) P(x) v = (1-n) Q(x)$$

Step III. Solve this linear ODE in v using the IF method.

Step IV. Back-substitute $v = y^{1-n}$ to get the answer in terms of y .

Worked Examples — Bernoulli Equation

Example 4: $\frac{dy}{dx} + y = xy^3$ $[n = 3]$

Step I. Divide both sides by y^3 :

$$y^{-3} \frac{dy}{dx} + y^{-2} = x$$

Step II. Substitute $v = y^{1-3} = y^{-2}$.

Differentiating: $\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx}$, so $y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$.

Substituting:

$$-\frac{1}{2} \frac{dv}{dx} + v = x$$

Multiply through by -2 to get standard linear form:

$$\frac{dv}{dx} - 2v = -2x$$

Step III. Solve the linear ODE. Here $P = -2$, $Q = -2x$:

$$\text{IF} = e^{\int -2 dx} = e^{-2x}$$

Multiply by e^{-2x} :

$$\frac{d}{dx}(v e^{-2x}) = -2x e^{-2x}$$

Integrate the right-hand side by parts. Let $u = x$, $dw = e^{-2x} dx$, so $w = -\frac{1}{2}e^{-2x}$:

$$\int -2x e^{-2x} dx = -2 \left[-\frac{x e^{-2x}}{2} - \int \left(-\frac{e^{-2x}}{2} \right) dx \right] = -2 \left[-\frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} \right] = x e^{-2x} + \frac{e^{-2x}}{2}$$

So:

$$v e^{-2x} = x e^{-2x} + \frac{e^{-2x}}{2} + C$$

Divide through by e^{-2x} :

$$v = x + \frac{1}{2} + C e^{2x}$$

Step IV. Back-substitute $v = y^{-2}$:

$$y^{-2} = x + \frac{1}{2} + C e^{2x}$$

Example 5: $\frac{dy}{dx} - \frac{y}{x} = \frac{y^2}{x^2}$ $[n = 2]$

Step I. Divide both sides by y^2 :

$$y^{-2} \frac{dy}{dx} - \frac{y^{-1}}{x} = \frac{1}{x^2}$$

Step II. Substitute $v = y^{1-2} = y^{-1}$.

Differentiating: $\frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$, so $y^{-2} \frac{dy}{dx} = -\frac{dv}{dx}$.

Substituting:

$$-\frac{dv}{dx} - \frac{v}{x} = \frac{1}{x^2}$$

Multiply through by -1 :

$$\frac{dv}{dx} + \frac{v}{x} = -\frac{1}{x^2}$$

Step III. Solve the linear ODE. Here $P = 1/x$, $Q = -1/x^2$:

$$\text{IF} = e^{\int 1/x \, dx} = e^{\ln x} = x$$

Multiply by x :

$$\frac{d}{dx}(vx) = x \cdot \left(-\frac{1}{x^2}\right) = -\frac{1}{x}$$

Integrate:

$$vx = -\ln|x| + C \implies v = \frac{-\ln|x| + C}{x}$$

Step IV. Back-substitute $v = y^{-1} = 1/y$:

$$\frac{1}{y} = \frac{C - \ln|x|}{x} \implies \boxed{y = \frac{x}{C - \ln|x|}}$$

Comparison: Linear vs Bernoulli

Linear ODE	Bernoulli Equation
$dy/dx + P(x)y = Q(x)$	$dy/dx + P(x)y = Q(x)y^n$, $n \neq 0, 1$
Already linear — use IF directly	Non-linear — divide by y^n , substitute $v = y^{1-n}$ first
$\text{IF} = e^{\int P \, dx}$	Same IF method, but applied to the reduced linear ODE in v
Solution in terms of y directly	Must back-substitute $v = y^{1-n}$ at the end

Summary

Concept	Key Point
Standard form	$dy/dx + P(x)y = Q(x)$ — coefficient of dy/dx must be 1
IF for linear ODE	$\text{IF} = e^{\int P(x) \, dx}$
LHS after multiplying	Always collapses to $d/dx [\text{IF} \cdot y]$
Bernoulli equation	$dy/dx + Py = Qy^n$, $n \neq 0, 1$
Bernoulli substitution	$v = y^{1-n} \Rightarrow dv/dx = (1-n)y^{-n} dy/dx$
Reduced equation	Linear ODE in v : $dv/dx + (1-n)Pv = (1-n)Q$
Back-substitution	Replace v with y^{1-n} at the very end

End of Lecture 7